Experiment 6: Write the python for Vacuum Cleaner Problem

Aim:

To write a Python program to implement the Vacuum Cleaner Problem using an intelligent agent.

Algorithm:

1. Initialize rooms A and B as dirty.
2. Place the vacuum cleaner in room A.
3. Check whether the current room is dirty.
4. If dirty, clean the room.
5. Move the vacuum cleaner to the next room.
6. Repeat until both rooms are clean.

Program:

room = {'A': 'Dirty', 'B': 'Dirty'}position = 'A'while 'Dirty' in room.values(): print("Room:", position) if room[position] == 'Dirty': print("Cleaning", position) room[position] = 'Clean' if position == 'A': position = 'B' else: position = 'A'print("All rooms are clean")

Output:

Room: A

Cleaning A

Room: B

Cleaning B

All rooms are clean

exp 6 vacume cleaner.py - C:/Users/yadal/OneDrive/Documents/AI/exp 6 vacume cleaner.py...

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```
room = {'A': 'Dirty', 'B': 'Dirty'}
position = 'A'

while 'Dirty' in room.values():
    print("Vacuum is in room", position)

    if room[position] == 'Dirty':
        print("Cleaning room", position)
        room[position] = 'Clean'

    if position == 'A':
        position = 'B'
    else:
        position = 'A'

print("Both rooms are clean")
```

IDLE Shell 3.14.7

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```
Python 3.14.7 (tags/v3.14.7:823f032, Aug  5 2026, 10:51:32) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/yadal/OneDrive/Documents/AI/exp 6 vacume cleaner.py ====
Vacuum is in room A
Cleaning room A
Vacuum is in room B
Cleaning room B
Both rooms are clean
>>>
```

Result:

Thus, the Python program for the Vacuum Cleaner Problem was executed successfully, and both rooms were cleaned.